

NOVEMBER 01 1998

Potential of microperforated panel absorber

Dah-You Maa



J. Acoust. Soc. Am. 104, 2861–2866 (1998)

<https://doi.org/10.1121/1.423870>



CrossMark

Export
Citation

Related Content

Numerical analyses of the sound absorption of cylindrical microperforated panel space absorbers with cores

J. Acoust. Soc. Am. (December 2015)

Sound absorption characteristics of a honeycomb-backed microperforated panel (MPP) absorber

J Acoust Soc Am (November 2006)

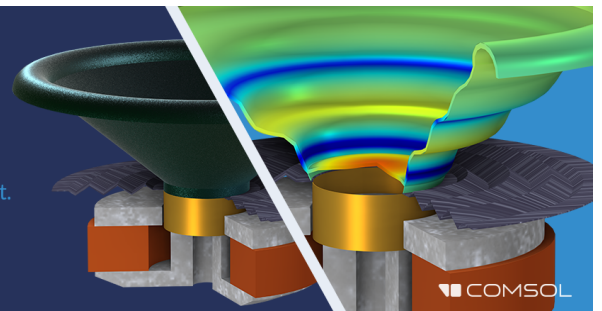
Nested sampling-based design of multilayer microperforated panel sound absorbers

Proc. Mtgs. Acoust. (June 2013)

Take the Lead in Acoustics

The ability to account for coupled physics phenomena lets you predict, optimize, and virtually test a design under real-world conditions – even before a first prototype is built.

» Learn more about COMSOL Multiphysics®



Potential of microperforated panel absorber

Dah-You Maa

Institute of Acoustics, Academia Sinica, P.O. Box 2712, Beijing 100080, People's Republic of China

(Received 7 November 1997; accepted for publication 16 July 1998)

Many applications have been found for the microperforated panel (MPP) absorber, on which the perforations are reduced to submillimeter size so that they themselves will provide enough acoustic resistance and also sufficiently low acoustic mass reactance necessary for a wide-band sound absorber. The most important parameter of the MPP is found to be the perforate constant k which is proportional to the ratio of the perforation radius to the viscous boundary layer thickness inside the holes. This, together with the relative (to the characteristic acoustic impedance in air) acoustic resistance r and the frequency f_0 of maximum absorption of the MPP absorber, decides the entire structure of the MPP absorber and its frequency characteristics. In other words, the MPP absorber may be designed according to the required absorbing characteristics in terms of the parameters k , r , and f_0 . Formulas and curves are presented toward this end. It is shown that the MPP absorber has tremendous potential for wide-band absorption up to 3 or 4 octaves and for low-frequency absorption with a cavity of depth small compared to the wavelength. Techniques of making minute holes (of 0.1–0.3 mm, say) have to be developed, though. © 1998 Acoustical Society of America. [S0001-4966(98)05710-5]

PACS numbers: 43.50.Gf, 43.55.Ev, 43.50.Jh [MRS]

INTRODUCTION

Perforated panels have been developed and used for some years,¹ but the necessary use of porous materials with it has reduced it to protective covering material.² Microperforated panels (MPP) were developed in the late sixties of the twentieth century, when a robust sound absorber was needed for severe environments, without fibrous, porous materials. An approximate theory was presented later.³ Unlike ordinary perforated panels where the perforations are in millimeters or even centimeters, with scarcely little inherent acoustic resistance, the perforations in MPP were reduced to submillimeter size (diameter 0.5–1 mm), so as to provide, by themselves, enough acoustic resistance and low acoustic mass reactance necessary for wide-band sound absorber, without additional fibrous, porous materials. Up to the present, the MPP absorber has usually provided sound absorption in a frequency band of one to two octaves, a wide band which no other resonator-absorber can do. But still, it is not quite enough for a general purpose sound absorber. Double resonators with two MPPs in tandem have been developed,^{4,5} and special design problems considered,⁶ to broaden the absorption band.

The MPP absorbers have a simple structure and absorption characteristics that are exactly predictable. The panel may be made of any material from cardboard, plastic, plywood to sheet metal, with any finishing or decoration to suit the purpose.⁷ The fact that the designs within the framework of the existing theory have limited absorption band is due to reasons not in the MPP absorber itself, but in the approximation made in the existing theory. It is the purpose of the present work to find the exact solution of the MPP equations in order to get a full view of the possibilities of the MPP absorber to facilitate better exploitation of the MPP absorbers.

It is found that the structure and frequency characteristics

of the MPP absorber depend on its relative acoustic resistance r , the resonance frequency f_0 , and the perforate constant $k = r_0 \sqrt{\rho_0 \omega / \eta}$ of the perforations, where r_0 is the radius of the perforations, ω the angular frequency, and η the coefficient of viscosity. The maximum absorption coefficient occurs at the resonance frequency f_0 and the possible band width is found to be given by the frequency interval (the ratio of the upper to lower half-absorption frequencies) $B = \pi / \cot^{-1}(1+r) - 1$, which may be very large when r is large. But the realization of the tremendous potential of the MPP absorber depends on the key parameter k . For a considerable range of k values, the maximum possible bandwidth becomes reality or near to reality. The absorption band narrows down quickly when k is increased beyond a certain value. But a small value of k calls for minute holes. Thus the perforate constant is essential for the MPP absorber design structure as well as characteristics.

I. THE MICROPERFORATED PANEL¹

The MPP may be considered a lattice of short narrow tubes, separated by distances much larger than their diameters, but small compared to the wavelength of impinging sound wave. The propagation of a sound wave in a tube was treated by Lord Rayleigh,⁸ and the treatment was simplified by Crandall⁹ for short tubes. The equation of aerial motion in a tube short compared to the wavelength is

$$j\omega\rho_0 u - \frac{\eta}{r_1} \frac{\partial}{\partial r_1} \left(r_1 \frac{\partial}{\partial r_1} u \right) = \frac{\Delta p}{t}, \quad (1)$$

where Δp is the sound pressure difference between the ends of the tube, t the length of the tube (equal to the thickness of the panel), ρ_0 the density of air, η its coefficient of viscosity, and r_1 the radius vector of cylindrical coordinates inside the tube. The equation may be solved for the particle velocity u , and the ratio of Δp to the average value of u over the cross-

sectional area of the tube gives the specific acoustic impedance of the short tube

$$Z_1 = \frac{\Delta p}{u} = j\omega\rho_0 t \left[1 - \frac{2}{k\sqrt{-j}} \frac{J_1(k\sqrt{-j})}{J_0(k\sqrt{-j})} \right]^{-1}, \quad (2)$$

where $k = r_0 \sqrt{\rho_0 \omega / \eta}$ (x had been used for this quantity in earlier papers,¹ it is changed here to avoid confusion), r_0 is the radius of the tube, J_1 the Bessel function of the first kind and first order, and J_0 same of zeroth order. It is seen that the quantity k is proportional to the ratio of the radius to the viscous boundary layer thickness inside the tube, and may be termed as the perforate constant. The limiting values of Z_1 for small and large values of k were given by Crandall as

$$Z_1 \rightarrow \frac{4}{3} j\omega\rho_0 t + \frac{32\eta t}{d^2}, \quad \text{as } k < 1, \quad (3a)$$

$$\rightarrow j\omega\rho_0 t + \frac{4\eta t}{d} \sqrt{\frac{\omega\rho_0}{2\eta}} (1+j), \quad \text{as } k > 10. \quad (3b)$$

These formulas have been used since Zwikker and Kosten¹ in the theory of absorbing materials. But the intermediate values of k between 1 and 10 are very important, especially for the MPP. An approximate formula,

$$Z_1 = \frac{32\eta t}{d^2} \left(1 + \frac{k^2}{32} \right)^{1/2} + j\omega\rho_0 t \left(1 + \left(3 + \frac{k^2}{2} \right)^{-1/2} \right), \quad (4)$$

has been developed to be valid for all k values by combining Eqs. (3a) and (3b) so that it becomes Eq. (3a) when k^2 terms in the brackets are omitted (k small) and Eq. (3b) when the k^2 terms are only retained (k large). The maximum error is only about 6% at some value of k between 1 and 10 from the exact value given by Eq. (2).

In Eq. (4) it is necessary to add the end corrections which have been discussed by Morse and Ingard.¹⁰ In his resonator paper,¹¹ Ingard suggested the use of values due to Rayleigh,¹² viz., the resistance due to air flow friction on the surface of the panel, $(1/2)\sqrt{2\omega\rho_0\eta}$ (the value used in earlier papers³ was too high, error is introduced when k is large), as the air flow is squeezed into the small area of the inlet end of the hole; and the mass reactance due to the piston sound radiation at both ends, $0.85d$. For normal incidence of sound wave on the panel, the wave motion in all the short tubes is in-phase and additive. The relative (to the characteristic impedance $\rho_0 c$ in air) acoustic impedance of the MPP is then

$$z = Z_1 / (\sigma\rho_0 c) = r + jx_m = r + j\omega m, \quad (5)$$

where

$$r = \frac{32\eta t}{\sigma\rho_0 c d^2} k_r, \quad k_r = \left[1 + \frac{k^2}{32} \right]^{1/2} + \frac{\sqrt{2}}{32} k \frac{d}{t}, \quad (5a)$$

$$\omega m = \frac{\omega t}{\sigma c} k_m, \quad k_m = 1 + \left[1 + \frac{k^2}{2} \right]^{-1/2} + 0.85 \frac{d}{t}, \quad (5b)$$

and

$$k = d \sqrt{\omega\rho_0/4\eta}, \quad (6)$$

where d is the orifice diameter, and t the panel thickness. [A useful design formula obtained from Eq. (6) is $k = d\sqrt{f/10}$,

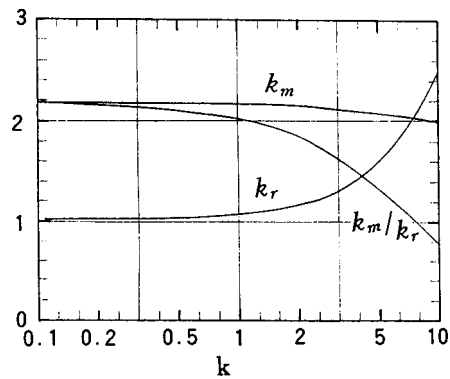


FIG. 1. Constants of MPP as functions of the perforate constant. k_r —resistance coefficient; k_m —mass reactance coefficient.

where d is specified in mm and f , in Hz.] The ratio d/t is usually nearly one and is taken so, small variations of the actual value are not important. The resistance coefficient k_r , mass reactance coefficient k_m , and their ratio k_m/k_r , important in the computation of r , m , and x_m , are plotted in Fig. 1 as functions of k . It is seen that k_m is practically constant, varying only by 2% in the range from 0.1 to 5 of k , but the resistance coefficient k_r increases with k significantly, especially when k is larger than 1. The maximum error of the approximate formula of acoustic impedance (5) is also 6% compared to the exact value from Eq. (2). Measurement techniques have been proposed for the orifice.^{10,13,14} Experiments¹⁵ agree well with the theoretical results (5) within 10% when the sound intensity is low, and further end corrections are necessary due to jet formation at the ends of the tubes,^{15,16} when the sound pressure level is well over 100 dB, depending on the perforation area ratio and resonance condition (about 100 dB at resonance when $\sigma = 0.01$). The end correction of the mass reactance, however, decreases with increasing intensity, perhaps due to the fact that the piston radiation at the ends of the narrow tube is partly blown away by the jet formed at the high intensity and it is proposed¹⁶ that the end correction of the acoustic mass should change the last term of k_m to $0.85(d/t)(1 + u_0/\sigma c_0)^{-1}$, where u_0 is the peak particle velocity inside the tube and σ the perforation ratio. The linear and nonlinear end corrections do not seem to be directly additive. Only the linear end corrections exist when the sound intensity is low, and the end corrections gradually change from linear values to nonlinear values after the particle velocity inside the tubes exceeds a certain value.

The perforation area ratio may be found from the resistance equation (5a) to be given by

$$\sigma/t = \frac{32\eta}{\rho_0 c} k_r / (rd^2). \quad (7)$$

The value of this ratio is important, while a small change of σ or t is usually allowed (t is ordinarily taken to be equal to d , because it is nearly so) the exact value is necessary here. Substituting this ratio into Eq. (5b), it yields the mass reactance

$$x_m = (k_m/8k_r)rk^2. \quad (8)$$

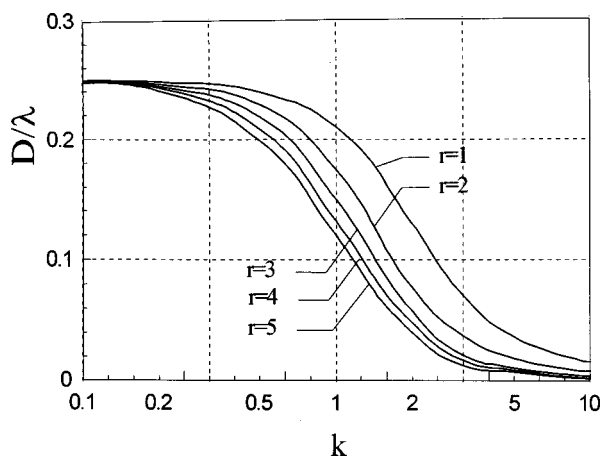


FIG. 2. Relative cavity depth D/λ as function of perforate constant k for different values of relative acoustic resistance.

This does not change the value of x_m ; it only relates x_m to the choice of r . The value of the ratio k_m/k_r is approximately 2 in the range of k from 0.5 to 2, which is the most important region in MPP design. Equation (8) may be taken as $x_m = 0.25rk^2$ in this region.

II. THE MPP ABSORBER

A microperforated panel fixed before a solid surface with a cavity of thickness D makes an MPP absorber. The relative acoustic resistance r and mass reactance $x_m = \omega m$ are given by Eq. (5) and the relative acoustic reactance of the cavity is $-\cot \omega D/c$. For normal incidence, the sound absorption coefficient is

$$\alpha = \frac{4r}{(1+r)^2 + (\omega m - \cot(\omega D/c))^2}. \quad (9)$$

Its maximum value

$$\alpha_0 = 4r/(1+r)^2 \quad (10)$$

occurs at the resonance frequency f_0 given by

$$\omega_0 m - \cot \frac{\omega_0 D}{c} = 0. \quad (11)$$

This gives

$$\frac{\omega_0 D}{c} = \cot^{-1} \omega_0 m \quad (12)$$

and the cavity depth ratio is obtained by a division with 2π

$$\frac{D}{\lambda} = \frac{1}{2\pi} \cot^{-1} \omega_0 m, \quad (13)$$

λ being the wavelength corresponding to the resonance frequency f_0 . Thus given the values of k , r and f_0 , the structure of the MPP absorber is completely determined. The cavity depth ratio as given by Eq. (13) is plotted in Fig. 2. It is 1/4 for all r values, when k is zero, it decreases linearly but slowly as k increases, with a rate proportional to r at first, and then the rates increase to a maximum and finally it levels off to a rather small value when k is large. This gives the possibility of a very thin absorber for low frequency. An

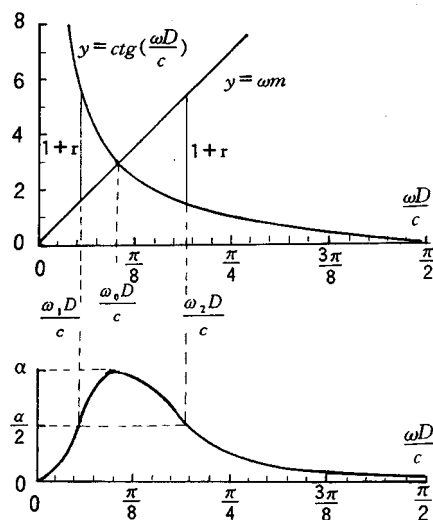


FIG. 3. Absorption band of MPP absorber.

important quantity is the ratio of ωm to $\omega_0 D/c$, or the ratio of the mass and cavity reactances at low frequency,

$$g = mc_0/D = \omega_0 m / \cot^{-1} \omega_0 m, \quad (14)$$

which is essentially a constant of the MPP, practically independent of frequency, and will prove to be important in the frequency characteristics of the MPP absorber.

The absorption coefficient α is one-half of the maximum value (10) at the frequencies given by

$$\omega m - \cot(\omega D/c) = \pm(1+r). \quad (15)$$

The resonance and half-absorption points are shown graphically in Fig. 3 of the ωm and $\omega D/c$ curves. In the figure, $\omega D/c$ is taken as the abscissa, which is proportional to the frequency. Thus the cotangent curve is valid for all designs, and the ωm line is nearly a straight line with a slope g given by Eq. (14). The lines intersect at the point of resonance, and their values differ by $1+r$ at half-absorption points. It is evident that the absorption band is wider as the value of $\omega_0 m$ is smaller, and also as r is larger. This is important when a wide-band MPP absorber is designed.

III. ABSORPTION BANDWIDTH

Equation (15) may be converted into an equation of the half-absorption frequencies,

$$\frac{\omega D}{c} = \cot^{-1} \left[\mp(1+r) + g \frac{\omega D}{c} \right], \quad (16)$$

with the help of Eq. (14), where the plus and minus sign is for the lower and higher half-absorption frequencies, respectively. The equation is quite simple, but an exact solution is not readily available. An approximate solution is, however, easy when g is small. From the derivative of the arc-cotangent function

$$\frac{d}{dx} \cot^{-1} \frac{x}{a} = -\frac{a}{a^2 + x^2}. \quad (17)$$

Equation (16) becomes, for the lower half-absorption frequency,

TABLE I. Absorption bandwidth possible of MPP absorber for different values of r ($k=0$, $D/\lambda=0.25$).

r	1	2	3	4	5
α_0	1	0.89	0.75	0.64	0.56
$(f_2 - f_1)/f_0$	1.41	1.59	1.69	1.75	1.79
f_2/f_1	5.78	8.76	11.82	14.91	18.02

$$\frac{\omega_1 D}{c} = \cot^{-1} \left[(1+r) + g \frac{\omega_1 D}{c} \right],$$

$$= \cot^{-1}(1+r) - \frac{1}{1+(1+r)^2} \cdot g \frac{\omega_1 D}{c}, \quad (18)$$

for low values of g . Solving

$$\frac{\omega_1 D}{c} = \cot^{-1}(1+r) \left/ \left[1 + \frac{g}{1+(1+r)^2} \right] \right. \quad (19)$$

to the first order of approximation. Similarly, for the upper half-absorption frequency,

$$\frac{\omega_2 D}{c} = [\pi - \cot^{-1}(1+r)] \left/ \left[1 + \frac{g}{1+(1+r)^2} \right] \right. \quad (20)$$

The frequency interval is the ratio of Eq. (20) to (19)

$$B = f_2/f_1 = [\pi / \cot^{-1}(1+r)] - 1, \quad (21)$$

which is constant to the first order of approximation, as g is small compared to unity. Thus the frequency interval of the absorption band of the MPP absorber is practically constant so far as rk^2 is smaller than 1. The possible values of B as well as the relative band width $F = (f_2 - f_1)/f_0$ are tabulated in Table I, for different values of r . As the absorption coefficient α_0 is not sensitive to the variation of acoustic resistance r , Eq. (21) as well as Table I demonstrate the possibility of exchanging bandwidth with absorption. The resonance frequency, however, varies with k all the way, as

$$\frac{\omega_0 D}{c} = \cot^{-1}(\omega_0 m) = \cot^{-1}[(k_r/8k_m)rk^2],$$

from Eqs. (8) and (11). It is $\pi/2$ when k is zero, and decreases slowly as k increases, exactly in the fashion of the cavity depth ratio (13), because the difference is only by a constant factor. It can be found that the resonance frequency is nearly the average of f_1 and f_2 when k is very small and gradually changes to their geometric mean when k is large.^{3,5} The resonance-absorption curve changes gradually from symmetrical to unsymmetrical, accordingly. The ratio of the resonance frequency to the lower half-absorption frequency is of interest, and may be found as

$$FR = \frac{f_0}{f_1} = \left(1 + r + \frac{g+1/3}{1+r} \right) \cot^{-1}(\omega_0 m), \quad (22)$$

if $g+1/3$ is small compared to $(1+r)^2$, $\omega_0 m$ is given by Eq. (8).

IV. ROLE OF THE PERFORATE CONSTANT k

Exact solution of Eq. (15) is desirable in order to have a view of the whole picture of the frequency characteristics of the MPP absorber. Numerical techniques were used to find

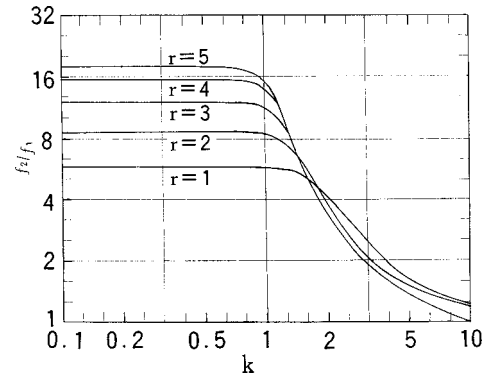


FIG. 4. Frequency interval of MPP absorption band as a function of the perforate constant k for different values of relative acoustic resistance.

the frequency interval in terms of r and k . The result is plotted in Fig. 4, showing the dependence of the frequency interval of absorption on the perforate constant k for different values of the acoustic resistance r . The magnitude of r decides the maximum possible bandwidth, but its realization is controlled by the value of k . The maximum possible values tabulated in Table I occur only when $k=0$. The values of B change little when k is small, reducing, for instance, only by 1% when k is increased to 0.9 (for $r=1$) or to 0.7 (for $r=5$). Shortly after these, the values of B drop rather steeply, more so for higher r , and the values for different r come closer and closer, at increasing values of k . The curves for r equal 2 and up cross each other between the values 1.25 and 1.6 of k , and the curve for $r=1$ comes slightly later. The order of the curves reverses after each crossing, and the curve for higher r lies lower, or the bandwidth is narrower for larger r . Thus, it is meaningless to take higher values of r if k is large. In other words, small values of k are essential for the wide-band MPP absorber. Larger values of k , however, are found important when r is small.

V. ABSORPTION IN DIFFUSE SOUND FIELD

The above sections are all for normal incidence of sound on the MPP absorber. In the case of oblique incidence, the MPP itself, as a locally reacting material, has its acoustic impedance $r + j\omega m$ unchanged. But in the cavity behind the panel, sound travels in direction θ to the normal, the same as the angle of incidence in the field, and the incident and reflected waves in the cavity have path difference $2D \cos \theta$ instead of $2D$ for normal incidence. Thus the relative acoustic impedance of the cavity becomes $(1/j \cos \theta) \times \cot((\omega D/c) \cos \theta)$. The relative acoustic impedance on a unit area of the panel surface for oblique-incident wave is thus

$$z = (r + j\omega m) \cos \theta - j \cot \left(\frac{\omega D}{c} \cos \theta \right),$$

and the absorption coefficient may be calculated as⁵

$$\alpha_0 = \frac{4r \cos \theta}{(1 + r \cos \theta)^2 + \left(\omega m \cos \theta - \cot \frac{\omega D}{c} \cos \theta \right)^2} \quad (23)$$

as the case discussed in Morse and Ingard.^{17,18} As a result, r , m , and D are all multiplied by the cosine. A multiplication of r by $\cos \theta$ means an increase of the absorption coefficient, if r is larger than one. The multiplication of m and D by the cosine is equivalent to a multiplication of ω by $\cos \theta$, as m and D of the MPP are fixed constants. Thus the resonance and the half-absorption frequencies are increased by the ratio $1/\cos \theta$, and the absorption band is shifted to higher frequencies. In a diffuse field, sound is incident at all angles on the MPP, and the absorption band extends continually to higher frequencies, and is broadened. Moreover, the cotangent curve is many branched, and the absorption band shown in Fig. 3 repeats itself when the argument is increased by π and its multiples, and there are many absorption bands along the frequency scale. The gap between the bands depends on the bandwidth of each band. The gaps between absorption bands will be, more or less, filled up if the bandwidth of each band is broad enough. Thus the absorption range of the MPP absorber is extended considerably in a diffuse sound field.

VI. DISCUSSION

It has been shown that the important parameters of the MPP absorber are its relative acoustic resistance r , determining its maximum absorption coefficient α_0 , the resonance frequency f_0 of the maximum absorption, and the perforate constant $k = r_0 \sqrt{\rho_0 \omega / \eta}$. These three parameters determine the structure and frequency characteristics of the MPP absorber.

The maximum absorption coefficient and maximum possible absorption bandwidth are decided by the value of r . An extremely wide band with good absorption is found to be possible with the MPP absorber, but its realization is limited by the value of the perforate constant. With the aid of Fig. 4, the values of k , r , and f_0 necessary to fulfill the frequency characteristics required of the absorber may be chosen. The resonance frequency is found from the frequency ratio (22). The desired frequency characteristics of the absorber are converted to a set of parameters. The design of the absorber may then be started from the absorption requirements of the absorber, expressed by the values of the parameters k , r , and f_0 . The perforation diameter required is found from Eq. (6) as

$$d = 2k \sqrt{\eta / \omega \rho_0}. \quad (24)$$

From Eq. (5a) of the relative acoustic resistance of the MPP, it may be solved by Eq. (7) for the perforation area ratio σ for panel thickness t , while k_r is given by Eq. (5a) and plotted in Fig. 1, taking $t/d = 1$. The exact value of t/d is not important for the value of k_r , unless it is far from 1, because it is in the correction term, and does not affect much. σ is the perforation area ratio and related to the ratio d/b as

$$\sigma = (\pi/4)(d/b)^2, \quad (25)$$

b being the separation of the perforations, if they are arranged in square lattices, and equivalently,

$$b = d \sqrt{\pi/4\sigma}. \quad (26)$$

Thus the constants of the microperforated panel are all determined, and the cavity depth ratio D/λ , which is given in

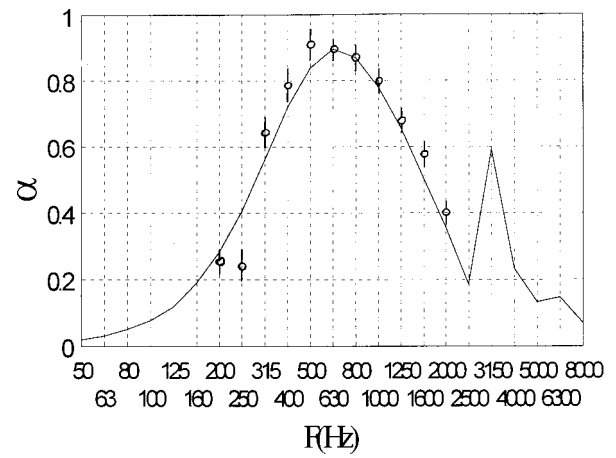


FIG. 5. Absorption curve of MPP absorber designed for 250–2000 Hz. $d = t = 0.2$ mm, $b = 2.5$ mm, $D = 0.06$ m, — theory, $\circ\circ\circ$ experimental points.

Eq. (13) and plotted in Fig. 2, determines the cavity depth. The mass reactance is given in Eq. (8), and the factor k_m/k_r , which is plotted in Fig. 1. Thus the design of the MPP absorber from the required k , r , and f_0 is simple and straightforward.

Take, for example, a general purpose sound absorber, an absorption band from 250–2000 Hz being desirable. This requires a frequency interval of 8. The family of f_2/f_1 curves in Fig. 4 indicate that this can be satisfied with a value of 1.25 for k and a value of r slightly over 2. The ratio of the resonance frequency to the lower half-absorption frequency for these k and r values is found from Eq. (22) as 3.05, and hence f_0 is to be 760 Hz. From these parameters, the constants of the MPP may be found as $d = t = 0.144$ mm, $\sigma = 0.52\%$, or $b = 1.77$ mm; the cavity depth formula (13) gives a value of 0.14 for D/λ , and D is 0.064 m. Thus the design is completed. An absorber with maximum absorption coefficient 0.88 and half-absorption between 250 and 2000 Hz is obtained. This calls for the making of minute holes, new techniques needed to be developed to realize the wide-band and high absorption possibility of the MPP absorber. Otherwise, some compromise must be made; if a lower resonance frequency is taken, for instance, $f_0 = 380$ Hz. The design will become $d = t = 0.2$ mm, $b = 2.5$ mm, and $D = 0.12$ m. The realization is easier, but the frequency range of absorption drops to 125–1000 Hz. The result is shown in Fig. 5, in which the solid line is the theoretical curve and the circles are obtained in measurements in a standing wave tube.

ACKNOWLEDGMENTS

This work has been supported by a grant from the National Natural Science Foundation of China. The figures were prepared by Professor Ke Liu.

¹C. Zwikker and C. W. Kosten, *Sound Absorbing Materials* (Elsevier, New York, 1949), Chap. II.

²L. L. Beranek, Ed., *Noise and Vibration Control* (McGraw-Hill, New York, 1971), p. 267.

³D.-Y. Maa, "Theory and design of microperforated-panel sound-absorbing construction," *Sci. Sin.* **XVIII**, 55–71 (1975).

- ⁴D.-Y. Maa, "Wide band sound absorber based on microperforated panels," *Proc. Inter-noise 84* (Honolulu, 1984), pp. 415–420; *Chin. J. Acoust.* **4**, 197–208 (1985).
- ⁵D.-Y. Maa, "Microperforated Panel wide-band absorber," *Noise Control Eng. J.* **29**, 77–84 (1987).
- ⁶D.-Y. Maa, "Design of microperforated panel construction," *Acta Acust.* **13**, 174–180 (1988); *Chin. J. Acoust.* **7**, 193–200 (1988).
- ⁷H. V. Fuchs and X. Zha, "Transparente Vorsatzschalen als alternative Schallabsorber im Plenarsaal des Bundestages," *Bauphysik* **16**, 69–80 (1994); "Einsatz microperforierter Platten als Schallabsorber mit inhärenter Dämpfung," *Acustica* **81**, 107–116 (1995).
- ⁸Lord Rayleigh, *Theory of Sound II* (MacMillan, New York, 1929 ed.), p. 327.
- ⁹I. B. Crandall, *Theory of Vibration System and Sound* (Van Nostrand, New York, 1926), pp. 229 et seq.
- ¹⁰P. M. Morse and U. Ingard, *Theoretical Acoustics* (McGraw-Hill, New York, 1968), pp. 460–463.
- ¹¹U. Ingard, "On the theory and design of acoustic resonators," *J. Acoust. Soc. Am.* **25**, 1044 (1953).
- ¹²Lord Rayleigh, Ref. 8, pp. 317–318, 487–491.
- ¹³M. R. Stinson and A. E. G. Shaw, "Acoustic impedance of small circular orifices in the plates," *J. Acoust. Soc. Am.* **77**, 2039–2042 (1985).
- ¹⁴D.-Y. Maa, "Measurements of acoustic impedences," in *Proc. ICA 12* (Toronto, 1986), M3-1.
- ¹⁵D.-Y. Maa, "Direct and accurate measurements of acoustic impedance of the microperforated panel," *Acta Acust.* (in Chinese) **8**, 257–261 (1983).
- ¹⁶D.-Y. Maa, "Microperforated panel at high sound intensity," *Proc. Inter-noise 94* (Yokohama, 1994), pp. 1511–1514; *Acta Acust.* **21**, 257–262 (1996).
- ¹⁷Reference 10, Eq. (6.3.8), p. 264.
- ¹⁸K. U. Ingard and H. Ising, "Acoustical nonlinearity of an orifice," *J. Acoust. Soc. Am.* **42**, 6–17 (1967).